EXPERIMENT-3

Experiment Title:

Design and Implementation of a 4-Bit Ripple Carry Adder Using Exclusive Gates (XOR) and Universal Gates (NAND/NOR) in Logisim.

Aim:

To design, simulate, and verify a 4-bit binary adder circuit using exclusive gates (XOR) and universal gates (NAND/NOR) in Logisim software, and to understand the working principle of ripple carry addition at the gate level.

Software Used:

Logisim (A graphical tool for designing and simulating digital logic circuits)

Theory:

Full Adder

A full adder adds three single-bit binary numbers (two inputs and a carry-in). Outputs: • Sum: $S = A \oplus B \oplus Cin$ • Carry Out: $Cout = (A \cdot B) + (Cin \cdot (A \oplus B))$

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

4-Bit Binary Adder (Ripple Carry Adder)

A 4-bit adder chains four full adders, with the carry output of each stage connected to the carry input of the next. This allows addition of two 4-bit binary numbers, producing a 4-bit sum and a carry out.

Universal Gates

NAND and NOR gates are called universal gates because any logic function can be implemented using only NAND or only NOR gates. The adder circuits can be constructed entirely from these gates.

Exclusive Gates

XOR and XNOR gates are particularly efficient for arithmetic operations. XOR directly implements the sum function, making adder design more compact.

Components Used:

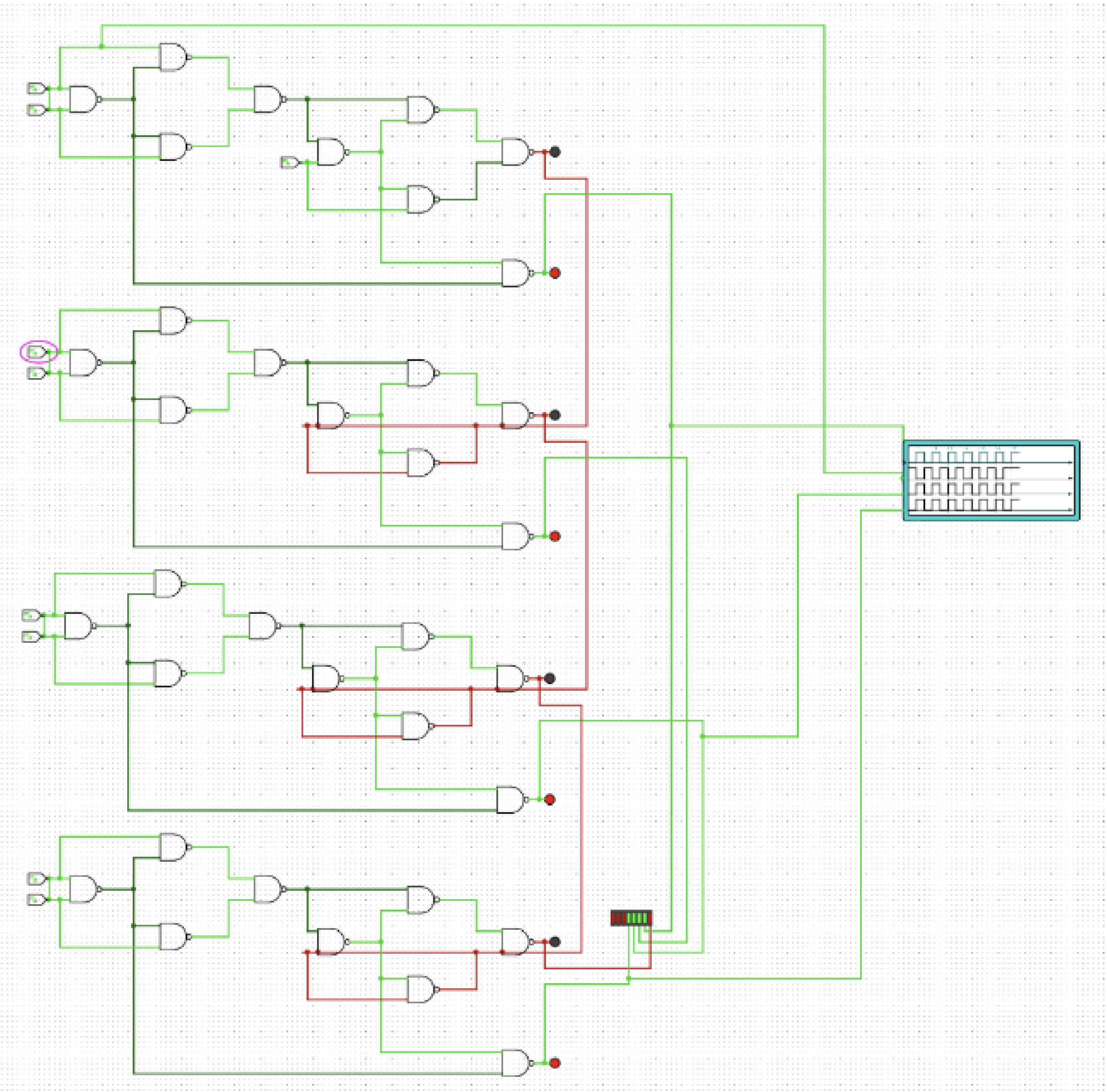
Component	Quantity
XOR Gate	8
AND Gate	8
OR Gate	4
NAND Gate	(as used)
NOR Gate	(as used)
Input Pins (A, B)	8
Output Pins (S)	4
Carry Output	1
Wires	As required

Procedure:

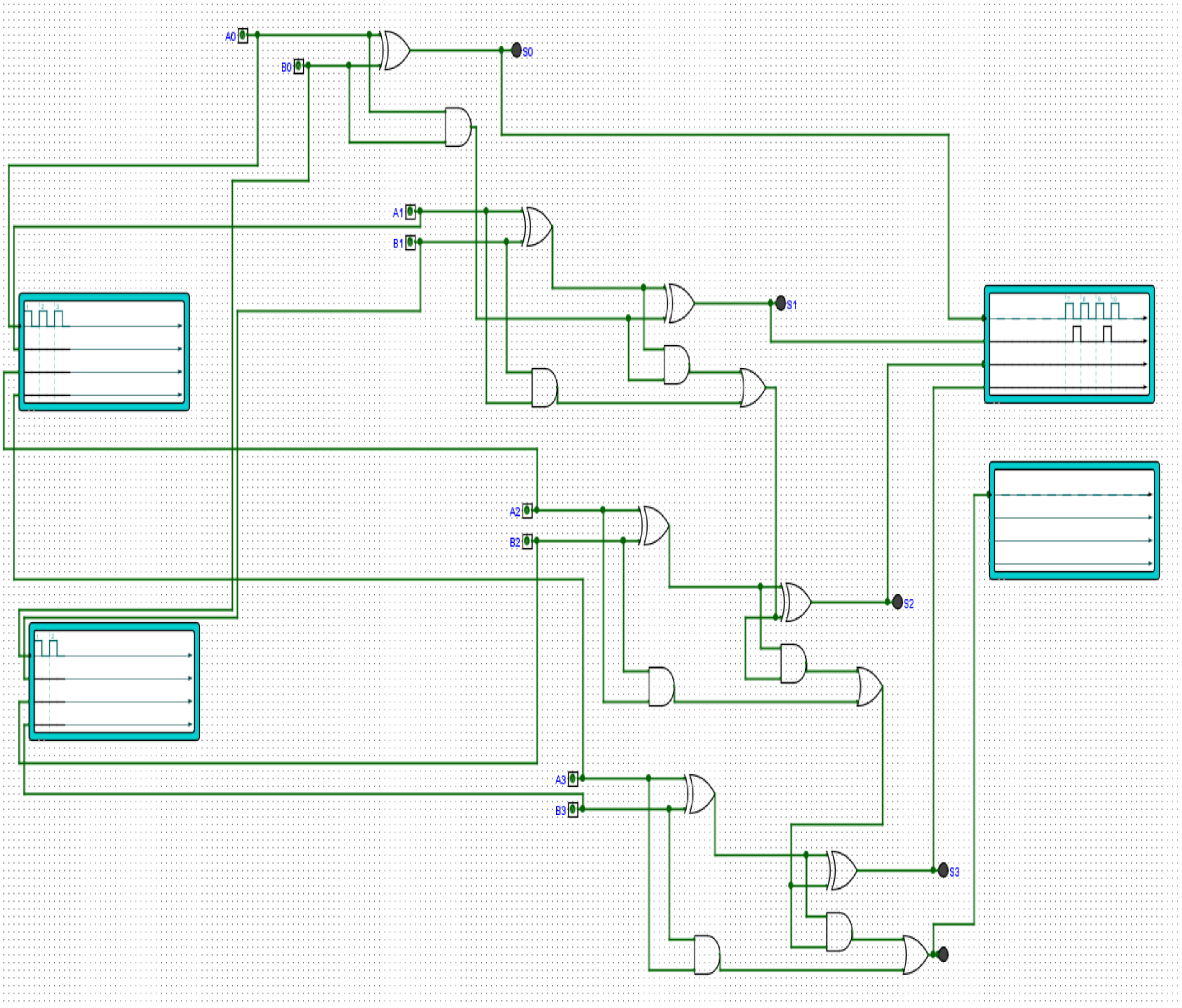
- Open Logisim and create a new project.
- Design a single-bit full adder using XOR, AND, and OR gates (or equivalent NAND/NOR implementation).
- Replicate the full adder module four times.
- Connect the carry-out of each adder to the carry-in of the next (ripple carry configuration).
- Label inputs as A0–A3 and B0–B3; label outputs as S0–S3 and Cout.
- Add waveform analyzers or probes to observe input/output signals.
- Apply various input combinations and verify the output against expected results.

Circuit Diagrams:

- Using universal gate - AND



- Using XOR & AND gates



Test Cases:

A (4-bit)	B (4-bit)	Expected Sum	Cout
0000	0000	0000	0
0001	0001	0010	0
0101	0011	1000	0
1111	0001	0000	1
1010	0101	1111	0
1111	1111	1110	1

Observations:

- The sum outputs (S0–S3) correctly reflected the binary addition of inputs A and B.
- The carry propagates correctly through each full adder stage.
- Waveform outputs (shown in screenshots) confirm expected timing and logic levels.

Result:

The 4-bit binary adder was successfully designed and simulated in Logisim using XOR and universal gates. All test cases produced correct sum and carry outputs, verifying the functionality of the ripple carry adder.

Conclusion:

- The experiment demonstrated the design of a combinational circuit for binary addition.
- XOR gates efficiently generate the sum, while AND/OR (or NAND/NOR) gates handle carry propagation.
- Understanding gate-level implementation reinforces concepts of digital arithmetic and lays the foundation for ALU design.

Author: Aayush Anand